



Takneek PS - Zenith

50 Points



Can He Master It All?

Introduction

In Greek mythology, Argus Panoptes was a giant with a hundred eyes, built such that some of the eyes were always open, a watchman that no threat could catch fully asleep. Every hospital could use one.

The diagnostic challenge lies in the sheer heterogeneity of the data. No two consecutive cases share a visual vocabulary, forcing a continuous mental reset across radically different scales of human biology, pivoting instantly from whole-organ anatomy down to microscopic cellular structures, each demanding an entirely different set of visual priors and clinical intuitions.

A model fluent in one of these can be functionally illiterate in front of another. Teaching a single system to be fluent in all of them at once, with no warning of which one it will encounter next, demands something of an Argus: a system with many eyes, each capable of seeing a different kind of world.

Problem Statement

Real hospitals don't keep a fellowship-trained specialist on standby for every modality, at every hour, in every ward. They run a small, tired rotation of generalists and an on-call pager and the queue doesn't stop growing just because it's 3 am. Your task is to build the specialist they don't have room to hire.

We need a medical imaging assistant that can take whatever image the grid hands it next, cross-sectional computed anatomical scans, magnetic resonance scans, projection radiographs, retinal photographs, dermatological surface images, microscopic cellular and tissue images, cross-sectional optical scans, or acoustic/sonographic images, in any order, with no indication of which type it is,



Takneek PS - Zenith

50 Points



and reason about what it shows. We need one system standing in for an entire missing medical department.

The system must be capable of handling the breadth of expertise required across these different forms of medical imagery, while presenting a **single unified interface** to the hospital.

Specifications:

Input

Your system will receive a collection of medical visual question-answering queries.

- Image: A medical image
- Question: A natural language question concerning the image
- Choices: Multiple choice options labelled A,B,C,D

A query may correspond to any of several medical imaging modalities. The system will not be told the imaging modality at inference time. The system must determine the answer using the provided image and question.

Your system must include a `predict()` function with the following requirements:

- Inputs: image, query, and choices.
- Output: A single character representing the predicted answer ('A', 'B', 'C', or 'D').
- `inference_time` is defined as the total execution time of this function for each query.
- Optimization: To minimize your `inference_time`, load all models before calling `predict()`. The function should be strictly limited to inference.

Output

For every query, the system must produce exactly one answer and inference time. The output must be saved in a CSV file with the following columns:

`query_id, answer, inference_time`

where:



Takneek PS - Zenith

50 Points



- `query_id` is the identifier supplied with the query
- `answer` must be exactly one of A,B,C,D which will be evaluated against ground truth
- `inference_time` is time in seconds taken to execute the `predict()` function in seconds

Example:

```
query_id,answer,inference_time
0001,B,22.73
0002,D,16.61
0003,A,25.47
```

Any malformed output, missing query, duplicate query, or answer outside {A,B,C,D} will be considered incorrect for that query.

Development Data:

A small development set containing 20 representative queries will be provided to all teams through a Google Drive link.

The development set will contain:

- medical images,
- questions,
- answer choices,
- query IDs.

The development set is intended to allow teams to verify their pipeline, benchmark their system, debug their implementation, and measure inference time.

The development set **should not be** assumed to represent the complete distribution of queries encountered during evaluation.

The development set will be released on **Wednesday, 26/08/2026**.



Takneek PS - Zenith

50 Points



Notes:

- The modality of an input image will **not** be provided to the system.
- The final system must produce exactly one answer (A, B, C, or D) for every query.
- Hard-coded answers for individual evaluation samples are strictly prohibited and will lead to test score 0/40.

Evaluation Process & Environment

Evaluation will be performed on a fresh Linux environment with one RTX 6000 Ada Generation(without any sudo access). All submissions will be evaluated under identical hardware and software conditions. Any failure during automated execution will not be manually debugged or corrected.

- **Environment & Weights:** All model weights should be contained inside the submission zip file, and zip file should handle inference directly and no training should take place on the evaluators side.
- **External Inference APIs: Use of external inference APIs or hosted AI services is strictly prohibited.** This includes, but is not limited to, OpenAI, Google/Gemini, Anthropic, Cohere, Mistral, and similar API-based or hosted inference services, regardless of whether the service is free or paid. **All inference must be performed locally on the provided evaluation hardware using the submitted code and locally available model weights.** No submission may transmit inputs to an external service for inference.
- **No Human Intervention:** The system must operate completely without human intervention once execution begins.

Execution Pipeline

To mimic a production deployment, the organizers will run the following commands to evaluate your submission.

```
unzip pool_bcs_zenith_takneek.zip
cd pool_bcs_zenith_takneek
```



Takneek PS - Zenith

50 Points



```
pip install -r requirements.txt
python eval.py --input_dir <path_to_queries>
```

The <path_to_queries> will have similar structure as a provided development set folder.

Output Format and Scoring

- **Output File:** The eval.py script must output a single CSV file named predictions.csv directly into the working directory.
- **Columns:** The CSV must strictly contain the columns query_id, answer, and inference_time.
- **Validity:** The answer column must contain exactly one of {A, B, C, D}. Any malformed output, missing queries, duplicates, or answers outside this set will be automatically marked as incorrect.
- **Final Score Calculation:** The score will be calculated as $\text{Score} = \text{Accuracy} - k * \text{Avg Inference Time}$. (k will be revealed with the development data)

Deliverables

Each team will submit 1 zip file named pool_bcs_zenith_takneek.zip (here bcs must be replaced by your pool name) containing all the following deliverables:

1. Make a **comprehensive report** for the methodology and pipeline used for the task (Maximum 15 pages excluding title, index & references). This should include,
 - a. Literature review
 - b. Idea and Methodology
 - c. Experimentation
 - d. Results
2. **Presentation**

Presentation should be submitted as a pdf, which will be presented after submission by the team members.

 - a. Maximum time to present: 10 minutes
 - b. A set of random questions will be asked to each member of the presenting team. Each question will be evaluated in a binary manner.
3. **Final Code**



Takneek PS - Zenith

50 Points



- README (Proper explanation of how to run the code, which scripts to use and all the components of the code)
- Final scripts containing the final system.
- Any dataset, which you may have felt the need to curate / references to publicly available datasets, if used.
- Any model weights needed to execute the testing script.

Scoring

- Technical Report - 30%
 - Literature review - 5%
 - Idea and Methodology - 10%
 - Experimentation - 5%
 - Results - 10%
- Presentation - 20%
 - Presentation - 10%
 - Q&A - 10%
- Model Evaluation - 50%
 - All code specifications and rules are followed as mentioned - 10%
 - Score on testing set - 40%
- Evaluation will consider **problem relevance, report comprehensiveness, clarity, model robustness, and novelty.**

Team Structure

Y24	1
Y25	3
Y26	2

Submission Deadline

2nd September 2026, 1159 hours